

# The Fabric Beneath Everything

*by Dustin Lee*

Let me start by telling you about the most embarrassing secret in all of physics.

We have two theories. Two extraordinarily successful, spectacularly well-tested, magnificent theories. One of them describes the very large — planets, stars, galaxies, the bending of light around black holes, the rippling of space itself when two neutron stars smash together. That theory is general relativity, Albert Einstein's masterpiece from 1915. It tells us that gravity is not a force pulling you toward the Earth. Gravity is the shape of space and time. Matter bends spacetime, and things move along the curves that result. It sounds crazy, but it works. It predicted the bending of starlight during a solar eclipse. It predicted that clocks tick slower near heavy objects — your GPS satellite has to correct for this every single day or it would drift by kilometers. It predicted gravitational waves — actual ripples in the geometry of the universe — which we finally detected in 2015 with the LIGO experiment, a hundred years after Einstein wrote down the equations.

The other theory describes the very small — atoms, electrons, quarks, the forces that hold your DNA together, the glow of your phone screen, the nuclear reactions that power the sun. That theory is quantum mechanics. It is, without any exaggeration, the most precisely tested theory in the history of science. Its

predictions for the magnetic properties of the electron agree with experiment to better than ten parts in a trillion. If you predicted the distance from New York to Los Angeles with that accuracy, you would be off by less than the width of a human hair. Quantum mechanics is not a rough sketch. It is a jewel of precision.

So we have two jewels. Two perfect, gleaming, astonishingly successful descriptions of nature. And here is the embarrassing part: they are fundamentally, profoundly, stubbornly incompatible with each other.

General relativity says that spacetime is a smooth, continuous fabric — you can divide it and subdivide it forever, and it stays smooth. It is dynamic, it bends, it ripples, it responds to the matter and energy within it. Matter tells spacetime how to curve, and spacetime tells matter how to move. It is a beautiful conversation between geometry and substance.

Quantum mechanics, on the other hand, treats spacetime as something completely different. In quantum mechanics, spacetime is a fixed stage — a rigid, unchanging backdrop against which the quantum drama plays out. Particles dance, wavefunctions evolve, probabilities interfere, but the stage itself just sits there, perfectly flat, perfectly inert, as if nothing interesting is happening to it at all.

Now, both of these things cannot be exactly right. Spacetime cannot be both a dynamic, bending participant and a rigid, fixed stage. And the trouble is not merely philosophical. There are real places in the universe — actual physical situations — where both theories must apply at the same time, and when you try to use them together, everything falls apart.

Think about the center of a black hole. General relativity tells us that matter collapses to a point of infinite density, a singularity where the curvature of spacetime blows up to infinity. But infinity is not a number — it is a confession that your theory has broken down. At that singularity, the region is so small and the energies so extreme that quantum mechanics must be relevant. But quantum mechanics does not know how to handle a spacetime that is curving so violently. It needs its nice flat stage. So both theories throw up their hands and say, "I cannot help you here."

Or think about the very beginning of the universe. Run the clock backward, and general relativity tells you the entire cosmos was once compressed to a single point — the Big Bang singularity, another place of infinite density and infinite curvature. But the entire universe compressed to subatomic scales is obviously a quantum situation. And again, neither theory can handle it alone, and they refuse to work together.

Or think about the Planck scale — a length so tiny that it makes an atomic nucleus look like a galaxy. It is about ten to the negative thirty-five meters. At that scale, the quantum fuzziness of spacetime itself becomes relevant. Spacetime should be fluctuating, bubbling, churning with quantum uncertainty. But general relativity says spacetime is smooth. And quantum mechanics says spacetime is fixed. Neither one can describe what is actually going on down there.

This is not a technical nuisance. This is a crisis. It means our picture of reality is fundamentally incomplete. We are like someone who has two beautifully detailed maps of the same city — but one shows the city as flat and the other shows it as hilly, and the street names do not match, and the rivers

are in different places. Both maps are incredibly useful for getting around. But they cannot both be perfectly correct descriptions of the same place.

And it gets worse. When physicists try the most obvious thing — taking the equations of general relativity and applying the standard rules of quantum mechanics to them, a procedure called "quantizing gravity" — the result is catastrophic. You get infinities. Not just one or two infinities that you might be able to sweep under the rug with some clever mathematics. You get infinities at every level of approximation, infinities that multiply and metastasize, infinities that cannot be absorbed or cancelled or tamed. The theory is, in the technical language, "non-renormalizable." It is not just difficult. It is nonsensical. The mathematical machinery that works so beautifully for electromagnetism and the nuclear forces simply explodes when you point it at gravity.

So unifying these two great theories is not optional. It is not a luxury project for physicists who have run out of other things to do. The universe itself demands it. There are real places and real situations where we need a theory that combines gravity and quantum mechanics, and we do not have one. The search for such a theory — a theory of quantum gravity, a unified picture — is arguably the deepest and most important problem in fundamental physics. It has been the great quest for nearly a century.

And the quest has not been for lack of trying. Let me give you a quick tour of the attempts so far, because understanding what has been tried — and what has not quite worked — is essential for appreciating the radical new idea I want to tell you about.

The most famous attempt is string theory. String theory says that the fundamental objects in the universe are not point particles but tiny, one-dimensional vibrating strings. Different vibrations of these strings produce different particles — one vibration mode gives you an electron, another gives you a quark, another gives you a photon. It is a beautiful idea, almost musical. And it naturally includes a vibration mode that behaves like a graviton — the hypothetical carrier of gravitational force — so it seemed like it might unify gravity with the other forces automatically.

But string theory demands something extraordinary: it only works mathematically if space has not three dimensions but nine or ten. The extra six or seven dimensions are supposedly curled up so small that we cannot see them. And the way those extra dimensions are curled up determines the physics we observe — the particle masses, the force strengths, everything. The problem is, there are an astronomically huge number of ways to curl them up. Roughly ten to the five hundredth power. That is a one followed by five hundred zeros. Each way gives you a different universe with different physics. String theory does not tell you which one is ours. After more than forty years of brilliant work by thousands of extremely clever people, string theory has not produced a single unique testable prediction. Not one. It has produced deep mathematical insights, beautiful connections, powerful tools. But as a physical theory of our universe, it remains frustratingly elusive.

Then there is loop quantum gravity. This approach takes a completely different tack. Instead of replacing particles with strings, it directly quantizes the geometry of spacetime itself. In loop quantum gravity, space is not a smooth continuum — it is a network, a mesh, a kind of chainmail made of discrete chunks of geometry. Area and volume come in tiny indivisible units,

like atoms of space. It is a remarkable picture, and it has had some real successes — for instance, it naturally provides a minimum length scale, which avoids the singularity problem. But it has its own struggles. It has difficulty recovering the smooth, continuous spacetime we observe at everyday scales — going from the atomic mesh to something that looks like Einstein's smooth geometry turns out to be technically very hard. And it does not include matter in any natural way. Matter has to be added by hand, separately. Geometry and matter remain distinct things that happen to coexist, rather than two aspects of a single underlying reality.

There are other approaches too. Causal set theory proposes that spacetime is fundamentally a discrete set of events with a built-in notion of before and after. Asymptotic safety suggests that gravity might be quantizable after all if you use the right mathematical framework. Causal dynamical triangulations try to build spacetime out of tiny triangular building blocks glued together. Each of these programs has contributed valuable insights. Each has illuminated some corner of the problem. But none has delivered the complete picture — a single, coherent framework that naturally gives you spacetime, matter, forces, quantum behavior, and gravity, all from one source.

Now here is where it gets interesting.

What if the reason none of these approaches has fully succeeded is that they are all starting in the wrong place? What if the very way we have been framing the problem contains a hidden assumption that is holding us back?

Think about what all these approaches have in common. String theory tries to quantize certain objects living in spacetime. Loop quantum gravity tries to quantize the geometry of spacetime. Causal sets try to discretize spacetime. In

every case, we start by treating spacetime and matter as separate categories of stuff — and then we try to find a theory that handles both. We try to quantize gravity, or geometrize quantum mechanics, or discretize the continuum — always as separate operations applied to things we have already assumed are distinct.

But what if they are not distinct? What if spacetime and matter are not two different things that need to be unified, but two different appearances of one single thing that was never divided in the first place?

This is the radical proposal I want to share with you. It comes from a line of thinking that says: stop trying to marry two incompatible theories. Start fresh. Begin with the simplest possible hypothesis — there is one thing, one underlying field, one substrate, and everything we observe is just different patterns and configurations of that one field.

Let me say that again because it is the key to everything that follows. There is a single substrate — let us just call it the *substrate* — and spacetime, particles, forces, gravity, quantum behavior, all of it, is just the substrate doing different things in different places and at different scales. Matter is not placed onto spacetime like furniture on a floor. Matter *is* spacetime, looked at differently. A particle is not a thing sitting in space. A particle is a knot in the fabric of space itself. And the smooth, gentle curvature we call gravity is what the fabric looks like when you step back and squint.

This is not a metaphor. I want to be very clear about that. This is a physical claim, with structural consequences, mathematical requirements, and testable predictions. It is not someone waving their hands and saying "everything is connected, man." It is a specific framework that tells you how

spacetime geometry emerges, why particles have the properties they do, where quantum uncertainty comes from, and what happens inside a black hole — all from one starting point.

But before we go further, let me tell you what the substrate is *not*, because this is where people almost always get confused.

The substrate is not an ether. Remember the luminiferous ether — that imaginary substance the nineteenth-century physicists invented to carry light waves, which Einstein showed does not exist? The ether was supposed to be a material substance sitting in space, providing a preferred reference frame, something you could measure your speed relative to. The substrate is nothing like that. The substrate does not sit in space. Space is a feature of the substrate's behavior. There is no space independent of the substrate for it to sit in. And the substrate does not provide a preferred frame of reference — the Lorentz symmetry we observe, the principle that no direction or velocity is special, emerges naturally from the substrate's collective behavior.

The substrate is also not a classical medium. It does not have a temperature or a viscosity or a molecular structure in the ordinary sense. It is not made of smaller things. It is where the explanatory chain ends. You might ask, "But what is the substrate made of?" And the answer is: that question does not have a deeper answer, just as asking "What is the electromagnetic field made of?" does not have a deeper answer. The electromagnetic field is not made of anything more fundamental — it is itself a fundamental thing. The substrate is like that, but even more fundamental, because the electromagnetic field is one of the things that emerges from it.



The substrate does not evolve in time, either — not in the usual sense. Time, like space, is not a pre-existing container that the substrate lives in. Time emerges from the causal structure of the substrate's dynamics. The substrate's internal relations have a natural ordering — some configurations lead to others, some states are causes and some are effects — and this causal ordering, when coarse-grained and averaged, is what we experience as the flow of time. Time is not the river that carries the substrate along. Time is the current that the substrate's own dynamics create.

So the substrate is physically real — as real as the electromagnetic field, as real as the gravitational field — but it is more fundamental than either. It is the thing out of which fields, particles, forces, space, and time all emerge. It is, if you like, the fabric beneath everything.

Now let me tell you about the deep principles that this framework rests on, and I will try to do it the way I think about it — not as a list of axioms but as a chain of insights, each one leading naturally to the next.

The first insight is the one we already discussed: matter and geometry are inseparable. They are not two substances interacting. They are two descriptions of one substance. Think about a vortex in the ocean. A whirlpool. Is the vortex a separate thing from the water? Can you remove the vortex and leave the water behind, or remove the water and keep the vortex? Of course not. The vortex *is* the water — it is the water in a particular configuration, a particular pattern of flow. Destroy the water, and the vortex disappears. Smooth out the flow, and the vortex is gone, but the water remains. The vortex and the calm water are not different substances. They are different behaviors of the same substance.

That is exactly the relationship between matter and spacetime in this framework. A smooth, gently curved region of the substrate is what we call geometry — the fabric of spacetime. A sharply localized, tightly wound, topologically nontrivial region of the substrate is what we call a particle. Same stuff. Different configurations. You do not put particles into spacetime. Particles are spacetime, twisted up.

The second insight is about gravity. In the standard picture, gravity is supposed to be carried by a force particle — the graviton, analogous to how electromagnetism is carried by photons. And when you try to build a quantum theory of gravity using gravitons, you get the horrible infinities I mentioned. But in the substrate framework, there is no fundamental graviton. Gravity is not a force transmitted by a particle. Gravity is the curvature response of the substrate. When a region of the substrate is tightly wound into what we call a particle, it modifies the substrate in its neighborhood. That modification ripples outward as a change in the large-scale geometry. Other configurations — other particles — deviate from straight-line paths because the geometry they are traveling through has been altered. That is gravity. No graviton needed. The graviton, if it appears at all, appears only as a useful fiction — an effective description that works well at low energies, much the way phonons in a crystal are useful fictions that describe sound waves without being fundamental objects.

Let me dwell on this for a moment, because it is important. Think about sound waves in air. At everyday scales, you can describe sound as a wave — it has a frequency, a wavelength, it diffracts around corners, it interferes with other sound waves. And if you quantize the sound wave, you get a phonon — a quantum of sound. Phonons are incredibly useful in solid-state physics. You

can calculate with them, scatter them off each other, build a whole theory around them. But nobody thinks phonons are fundamental. Nobody thinks that at the deepest level of reality, sound is made of tiny sound particles. We know that sound is a collective vibration of atoms, and the phonon is just a convenient mathematical wrapper for that collective behavior. If you push the phonon description to very high energies or very short wavelengths — down to the scale of individual atoms — the phonon description breaks down. It starts giving you nonsense. But that is not a crisis. It just means you have reached the scale where you need to talk about atoms instead of phonons.

The graviton is in exactly the same situation. It is a phonon-like effective particle that describes the collective behavior of the substrate at long distances and low energies. At those scales, it works fine. You can use graviton language to describe gravitational waves, and you will get the right answers. But at very high energies — near the Planck scale, inside black holes — the graviton description breaks down, just as phonon physics breaks down at the atomic scale. And the resolution is the same: do not panic, do not add epicycles, do not invent new particles. Just recognize that you have reached the scale where you need to talk about the fundamental substrate instead of its collective excitations.

This, by the way, is exactly why perturbative quantum gravity fails. When you write down Feynman diagrams with virtual gravitons running in loops, and you find divergences at every order, the conventional reaction is to say the theory is sick and needs new physics to fix it. But the substrate framework gives a different diagnosis. The divergences are not a sign that the theory needs fixing. They are a sign that you are using the wrong description. You are trying to describe a fundamentally non-perturbative, collective phenomenon

using perturbative, particle-based language. It is like trying to describe a tsunami by tracking individual water molecules. The description is not wrong in principle — every molecule does contribute — but in practice, it is the wrong level of description, and it produces nonsense.

The third insight is perhaps the most surprising: Heisenberg's uncertainty principle, that iconic statement that you cannot simultaneously know a particle's exact position and exact momentum, is not a brute fact about measurement. It is not a limitation of our instruments or our cleverness. It is a geometric constraint built into the substrate itself.

Here is how to think about it. Position and momentum are not independent properties. They are complementary descriptions of the same underlying substrate configuration. Position tells you the spatial profile — how sharply localized the configuration is, where the knot in the fabric sits. Momentum tells you the gradient structure — how rapidly the configuration varies from point to point, how much it is oscillating.

Now think about a wave. Imagine a wave on the ocean. If you want the wave to be concentrated in one tiny spot — sharply localized — then it has to have very steep gradients. The water has to rise and fall very quickly in a very small region. That means the wave contains many different wavelengths, all superimposed. Its frequency content is spread out. On the other hand, if you want a wave with a single, pure frequency — a perfect sine wave — it has to extend over a very large region of ocean. It cannot be localized at all. A wave that is narrow in space is necessarily broad in frequency, and a wave that is narrow in frequency is necessarily broad in space. This is not quantum mechanics. This is a mathematical fact about waves, discovered by Joseph Fourier two hundred years ago.

The Heisenberg uncertainty principle is this same tradeoff, applied to the substrate. A substrate configuration that is sharply localized in position necessarily has a broad distribution of gradients — which means uncertain momentum. A configuration with well-defined gradients — definite momentum — is necessarily spread out in position. The substrate cannot be simultaneously narrow in both descriptions, for the same reason an ocean wave cannot be simultaneously a sharp spike and a pure tone.

But here is what the substrate framework adds to this story. The specific size of the minimum uncertainty — the exact quantitative bound, which involves Planck's constant — is determined by the information density of the substrate. The substrate has a finite amount of information per region. Any localized configuration engages a finite number of degrees of freedom, and those degrees of freedom have a finite information budget. You can spend that budget on precise position or precise momentum, but you cannot overspend. The minimum uncertainty product — which *is* Planck's constant — is set by how much information the substrate can hold per unit of its own internal structure.

Think about what this means. Planck's constant — that mysterious number that appears in every quantum equation, that constant we have measured to exquisite precision but never explained — is not a magical decree from the universe. It is a structural parameter, determined by the internal architecture of the substrate. It is like the speed of sound in a material. The speed of sound in air is not a fundamental constant of nature. It is determined by the mass of air molecules, the temperature, the pressure. Change the medium, change the speed of sound. In the same way, Planck's constant is determined by the density and connectivity of the substrate. A hypothetical universe with a

different substrate structure would have a different Planck's constant. Ours has the value it has because our substrate has the particular structure it has.

And once you see uncertainty as a geometric property of the substrate, rather than a free-floating axiom of quantum mechanics, you immediately get new predictions. If spacetime is emergent and uncertainty is geometric, then the uncertainty bound should be modified by curvature. Near a black hole, where the substrate is intensely curved, the information density is enhanced — more degrees of freedom are packed into a given region. This tightens the uncertainty bound, and the result is phenomena like Hawking radiation and the Unruh effect — effects that already have strong theoretical support and that the substrate framework produces as direct consequences rather than separate discoveries. At cosmological scales, the expansion of the universe dilutes the information density, loosening the uncertainty bound, and this helps explain why quantum effects become negligible at large scales — why the universe looks classical when you are big enough. And at the Planck scale itself, the discrete microstructure of the substrate introduces additional uncertainty beyond the standard Heisenberg bound — what physicists call the generalized uncertainty principle. In the substrate framework, this is not an ad hoc modification. It is a natural consequence of the substrate having a finite granularity.

The fourth great insight concerns what holds spacetime together. What makes distant points part of the same universe? What is the glue?

The answer is entanglement.

Quantum entanglement — that strange correlation between distant particles that Einstein famously called "spooky action at a distance" — is not

just a curious quantum phenomenon. It is the structural fabric of spacetime itself. Substrate degrees of freedom are entangled with each other across extended regions. When two regions are highly entangled, they are geometrically close — nearby in the emergent spacetime. When two regions have little or no entanglement, they are geometrically far apart, or even disconnected — literally in different spaces.

This is an idea that has been growing in physics for over a decade. Mark Van Raamsdonk, a physicist at the University of British Columbia, argued in 2010 that spacetime is built from entanglement. He showed, using the tools of a correspondence between gravity and quantum mechanics called holographic duality, that if you reduce the entanglement between two regions of a quantum system, the corresponding spacetime geometry pulls apart. The two regions separate, the bridge between them narrows, and eventually, if you remove the entanglement entirely, the spacetime pinches off into two disconnected pieces. Without entanglement, no spacetime.

The substrate framework takes this idea and makes it concrete. Entanglement is not just associated with spacetime connectivity — it is the mechanism that generates it. Spacetime is the way highly entangled substrate degrees of freedom look from far away. And this immediately gives you one of the most important results in quantum gravity: the Bekenstein-Hawking entropy formula.

Let me tell you about this formula, because it is a key piece of the puzzle. In the 1970s, Jacob Bekenstein and Stephen Hawking discovered something astonishing about black holes. A black hole has entropy — it has a measure of how much disorder, how much hidden information, it contains. And the entropy is proportional not to the volume of the black hole, as you might

naively expect, but to the area of its event horizon — the surface of no return. This was deeply mysterious. Why area, not volume? In every other physical system, entropy scales with volume. A box of gas has more entropy if you make it bigger in all three dimensions. But a black hole's entropy cares only about its surface.

In the substrate framework, this is natural. A black hole is a region where the substrate is maximally curved, extremely tightly wound. The event horizon is the boundary between the interior and exterior substrate configurations. The entanglement entropy — the entropy that comes from the quantum correlations between interior and exterior degrees of freedom — is determined by how many substrate degrees of freedom straddle that boundary. And for a locally interacting system, which the substrate is, the number of degrees of freedom straddling a boundary is proportional to the area of the boundary, not the volume it encloses. This is called the "area law" for entanglement entropy, and it is a well-established result in quantum information theory. So the Bekenstein-Hawking formula is not a mysterious coincidence. It is a direct consequence of the substrate being a locally interacting quantum system. The microscopic degrees of freedom responsible for black hole entropy are identified: they are the substrate configurations straddling the horizon.

Now here is where the plot really thickens. The fifth insight — and maybe the deepest one — is about what the substrate framework does to the measurement problem in quantum mechanics.

The measurement problem is, I think, the most philosophically disturbing issue in all of physics. Quantum mechanics says that before you measure a particle, it does not have a definite position. It exists in a "superposition" — a combination of all possible positions at once. Then you measure it, and



suddenly, instantaneously, it "collapses" to a single definite position. But quantum mechanics never explains how or why this collapse happens. The equations of quantum mechanics — the Schrödinger equation — describe smooth, continuous, deterministic evolution. There is no collapse in the equations. Collapse is added by hand, as a separate postulate, a *deus ex machina* invoked every time a measurement occurs. And what counts as a "measurement"? The theory does not say. It is, as the great John Bell put it, "unprofessionally vague."

The substrate framework dissolves this problem entirely. In the substrate picture, evolution is strictly unitary. Always. No exceptions. No collapse. A measurement is just a physical interaction between two substrate configurations — the system you are measuring and the apparatus you are measuring it with. When they interact, entanglement develops between them. The information about the system's state gets distributed into the correlations between the system and the apparatus — and then into the correlations between the apparatus and its environment, and the environment and its environment, spreading outward into an astronomically large number of substrate degrees of freedom.

This is decoherence. The information is not destroyed. It is not "collapsed." It is redistributed, spread so thin across so many degrees of freedom that no local observer could ever gather it all back together. From the perspective of the observer — who is herself a substrate configuration, embedded in the same substrate — the outcome looks definite. It looks like collapse. But fundamentally, the full quantum superposition is still there. It has just become unobservable to any subsystem embedded within the substrate.

Suppose you measure whether a particle goes left or right. Before the measurement, the substrate configuration includes both possibilities. After the measurement, the observer's substrate configuration has become entangled with the particle's configuration in such a way that each branch of the superposition includes a correlated observer — one who saw "left" and one who saw "right." These branches rapidly decohere because the entanglement spreads into the surrounding substrate. To each observer within her branch, the outcome looks sharp and definite. No collapse is needed. No new postulates. No mysterious intervention by consciousness or measurement devices. Just the ordinary unitary evolution of one substrate, following one set of equations.

This also explains, in a satisfying way, why measuring position disturbs momentum and vice versa. It is not some mystical quantum weirdness. It is the finite information budget of the substrate at work. When you measure position, the interaction sharpens the spatial profile of the configuration — but the finite budget means that sharpening the spatial profile necessarily broadens the gradient structure. Information is redistributed, not created. You cannot gain precise knowledge of position without losing knowledge of momentum, because they are complementary uses of the same limited information resource.

All right. Now let me take you through something truly wonderful — the story of how spacetime itself emerges from the substrate. Because this is the heart of the whole framework, and it is a beautiful piece of logic.

Imagine — and this is an analogy, but a very good one — imagine a vast ocean. Not our ocean, with water and waves and fish, but a mathematical ocean: an enormous collection of substrate degrees of freedom, each one

interacting with its neighbors, each one carrying some internal information. There is no space yet. There are no distances. There is no geometry. There is just this enormously complex network of interacting degrees of freedom.

Now suppose you zoom out. Way out. You stop trying to track each individual degree of freedom and instead start looking at averages. You group the degrees of freedom into blocks — clusters of many individual elements — and you ask: what are the average properties of each block? How are the blocks connected to each other? How much information flows between them?

This is the process physicists call "coarse-graining," and it is exactly what happens when you go from a microscopic description to a macroscopic one. No individual water molecule has a temperature. Temperature is not a property of a single molecule. It is an average property that emerges when you look at billions of molecules together. No individual air molecule has pressure. Pressure is a collective property. You cannot point to any single molecule and say, "Here — this one is the pressure." It emerges from the statistics.

The same thing happens with spacetime. No individual substrate degree of freedom has "distance" associated with it. There is no tiny little ruler attached to each one. Distance, curvature, dimensionality — these are collective properties that emerge from the statistics of the substrate when you coarse-grain.

The emergence happens in three stages. In the first stage, you do the initial averaging. You group the substrate degrees of freedom into blocks and replace the detailed microstate of each block with a few averaged quantities. The connectivity pattern between blocks — which blocks are strongly correlated with which other blocks — becomes the seed of what will eventually

be a metric, a measure of distances. The localized excitations within blocks — the knots and twists — become the seed of what will eventually be energy and momentum. The internal entanglement patterns become the seed of what will eventually be quantum correlations.

In the second stage, you take the continuum limit. As the blocks get large compared to the Planck length but remain small compared to everyday scales, the discrete block data smooths out into a continuous geometry — a four-dimensional pseudo-Riemannian manifold, which is the mathematical object that general relativity uses to describe spacetime. And here is a remarkable thing: the number of dimensions is not assumed. It is derived. The substrate's internal connectivity structure determines the effective dimensionality after coarse-graining. It turns out that three spatial dimensions and one time dimension — what physicists call three-plus-one — is the stable configuration. Higher-dimensional configurations are dynamically unstable — they collapse or fragment. Lower-dimensional configurations cannot support the propagation modes that matter needs. Our universe has three spatial dimensions not because anyone decreed it, but because three is the number that survives.

Even the distinction between space and time — the Lorentzian signature, the fact that time is fundamentally different from space — emerges. The substrate's dynamics have an inherent asymmetry: evolution proceeds along one axis while exhibiting spatial extensiveness along the other three. This asymmetry, after coarse-graining, becomes the distinction between the one time dimension and the three spatial dimensions. Time is not just another dimension with a minus sign. The minus sign reflects a real, deep asymmetry in how the substrate behaves.

In the third stage, you derive the effective dynamics. You integrate out all the microscopic degrees of freedom — all the fine-grained detail you cannot see at macroscopic scales — and you ask: what equations do the coarse-grained, averaged quantities obey? And the answer, at leading order, is Einstein's field equations.

Let me tell you how this works, because it is one of the most beautiful arguments in all of theoretical physics, and the substrate framework gives it a solid foundation.

In 1995, Ted Jacobson, a physicist at the University of Maryland, published a remarkable paper. He showed that you could *derive* Einstein's field equations from thermodynamics. Not from geometry. Not from the action principle of general relativity. From thermodynamics — from the relationship between heat, entropy, and temperature.

Here is the logic. Consider a tiny patch of spacetime — what physicists call a "causal diamond," a small region bounded by light rays. If you are an accelerated observer hovering just inside the boundary of this region, you see a horizon — a surface beyond which you cannot see. And Unruh's effect tells you that the vacuum looks like a thermal bath to an accelerated observer. The horizon has a temperature.

Now, Jacobson assumed that the entropy of this horizon is proportional to its area — the Bekenstein-Hawking relation. And he applied the Clausius relation — the fundamental thermodynamic identity that says the heat flowing through a surface equals the temperature times the change in entropy. The constraint you get from demanding that this relation hold for every little causal

diamond, at every point of spacetime, in every direction — that constraint *is* the Einstein field equation.

Einstein's equation, in this view, is not a fundamental law. It is an equation of state — like the ideal gas law, which tells you the relationship between pressure, volume, and temperature for a gas. The ideal gas law is not a fundamental law of physics. It is a macroscopic relationship that emerges from the statistical behavior of billions of molecules. Einstein's equation is exactly the same kind of thing — a macroscopic relationship that emerges from the statistical behavior of enormous numbers of substrate degrees of freedom.

This is extraordinary. It means that gravity — the force that governs the motion of planets, the expansion of the universe, the bending of light — is an entropic effect. The substrate is "trying" to maximize its entanglement entropy, subject to the constraints imposed by energy conservation. The resulting geometry, the curvature of spacetime, is what you get when the substrate finds the most probable configuration. Gravity is not a fundamental force. It is a statistical tendency, like the tendency of gas to expand into an empty room.

And the substrate framework provides something that Jacobson's original derivation was missing: the microscopic foundation. Jacobson assumed the entropy-area relation. The substrate framework derives it from the area law for entanglement entropy in a locally interacting system. Jacobson assumed the Clausius relation. The substrate framework derives it from the unitary dynamics of the substrate. Even Newton's gravitational constant — that fundamental number that tells you how strong gravity is — is derived. It is determined by the density of substrate degrees of freedom and the strength of

their mutual entanglement. It is not a free parameter. It is a calculable quantity, like the speed of sound in a specific material.

Now let me tell you about where classical geometry breaks down, because this is where the substrate framework really earns its keep.

There are three regimes where Einstein's smooth geometry fails, and in each case, the substrate framework tells you what actually happens.

First, the Planck scale. At the tiniest distances — around ten to the negative thirty-five meters — the coarse-graining procedure that produces smooth geometry breaks down. The blocks become comparable to the individual degrees of freedom. You can no longer meaningfully average. This is where you must abandon the geometric description entirely and work with the fundamental substrate. The smoothness of spacetime is an approximation — an extraordinarily good approximation at everyday scales, but an approximation nonetheless. At the Planck scale, spacetime is not smooth. It is not anything that can be described in the language of geometry. It is pre-geometric — raw substrate, with its own rules and its own behavior.

Second, black hole interiors. General relativity says the interior of a black hole terminates in a singularity — a point of infinite curvature and infinite density. But infinity is a warning sign, not a physical reality. In the substrate framework, there is no singularity. As matter collapses under its own gravity, the substrate becomes more and more tightly wound. But the substrate does not permit infinite curvature, because it has a minimum length scale — the scale at which the individual degrees of freedom live. You cannot compress the substrate below this scale. What general relativity calls a singularity is actually a region where the substrate is maximally compressed, where the distinction

between geometry and matter dissolves entirely. The substrate is still there, still well-behaved, still evolving unitarily. It is just no longer meaningful to describe it in the language of smooth geometry.

Third, the very early universe. Run the clock backward, and the conventional picture gives you a Big Bang singularity — the entire universe compressed to a point. The substrate framework replaces this with something far more interesting: a pre-geometric phase. Before the "Big Bang," there was no spacetime. There was no metric, no distance, no curvature. There was the substrate, in a configuration so different from what we know that the concepts of space and time simply did not apply. Then something happened — a phase transition, like water freezing into ice or, more aptly, a gas condensing into a liquid. The substrate underwent a transition from its pre-geometric phase to its geometric phase. Spacetime condensed out of the substrate, like a crystal forming in a cooling liquid. What we call the Big Bang is this phase transition. What we call the rapid inflationary expansion that followed is the violent, rapid growth of the newly formed geometric phase as it spread and stabilized.

Let me turn now to one of my favorite parts of this whole story: particles.

In the substrate framework, a particle is not a fundamental object. It is not an irreducible, structureless point that exists as a primitive element of reality. A particle is a topologically stable configuration of the substrate — a knot in the fabric.

Think about what that means. Imagine a smooth sheet of rubber. You can stretch it, bend it, deform it in all sorts of ways, and it will spring back. A smooth configuration can relax to its lowest-energy state. There is nothing holding the deformation in place. But now imagine you twist the rubber sheet



and tie it in a knot. The knot is stable — not because some force is holding it together, but because of topology. You cannot untie a knot by smoothly deforming the sheet. You would have to cut it. The knot is trapped by its own structure.

Particles in the substrate framework are exactly like this. They are regions where the substrate has been twisted, wound, knotted in a way that cannot be smoothly unwound. The topology of the configuration prevents it from relaxing to the smooth ground state. And this topological stability is what gives particles their permanence. An electron does not decay because there is no lighter configuration with the same topological structure. There is nowhere for it to go. It is stuck — beautifully, permanently stuck — in its topological class.

And here is one of the most satisfying things about this picture: it explains why all electrons are identical. This is one of those facts about nature that we usually just accept without question, but when you stop and think about it, it is deeply strange. Every electron in the universe has exactly the same mass, exactly the same charge, exactly the same spin. Not approximately the same — *exactly* the same. How? Why? In the standard picture, this is simply asserted. Electrons are identical because that is what the theory says. There is no deeper explanation.

But in the substrate framework, the explanation is beautiful and natural. Topological invariants are integers. A knot is either a trefoil or it is not. It either has three crossings or it does not. There are no almost-trefoils, no 3.7-crossing knots. Topology is discrete even when the underlying medium is continuous. Two electrons are identical because they are the same topological class of substrate configuration. They have to be identical, for the same reason that all trefoil knots are identical regardless of whether they are tied in rope, in wire, or

in silk. The material does not matter. The topology does. And since the quantum numbers of particles are topological invariants, all particles of the same type must have exactly the same quantum numbers. There is no room for variation.

Now let me tell you how the specific quantum numbers arise, because this is genuinely wonderful.

Electric charge, in this framework, is a winding number. Imagine the substrate has internal degrees of freedom — not spatial dimensions, but internal properties at each point. These internal degrees of freedom can wrap around an internal cycle, like a path wrapping around a circle. The number of times they wrap — once, twice, three times — is an integer. It has to be an integer, because you either go all the way around or you do not. There is no such thing as wrapping one and a half times around a circle and ending up back where you started. This winding number is the electric charge. And because it is an integer, charge is quantized — it comes in discrete units. The electron has charge negative one. The proton has charge positive one. You never see a particle with charge one-third or two-sevenths. Charge quantization, which in the standard model is simply an observed fact without deep explanation, becomes a mathematical necessity.

But wait — what about quarks? Quarks have charges of plus two-thirds and minus one-third. Those are not integers. How does that fit? It fits because the internal space in which the winding occurs has a three-fold symmetry — reflecting the  $SU(3)$  structure of the strong force. In a space with three-fold symmetry, the winding can come in thirds. But — and this is crucial — only combinations that add up to integers can exist as free particles. A quark, with its fractional charge, cannot exist alone. It must always be bound with other

quarks in a combination whose total winding number is an integer. And this is exactly what we observe: quarks are always confined inside protons, neutrons, and other hadrons, never seen in isolation. Confinement is not just a dynamical accident. It is a topological necessity.

Color charge — the charge associated with the strong nuclear force — corresponds to orientation in the internal space. There are three independent orientations, which is why there are three colors. And the topological constraint that net color cannot exist alone is the deep reason for color confinement.

Spin — the intrinsic angular momentum of a particle — corresponds to the rotational structure of the localized configuration. A spin-one-half particle is one where the substrate configuration has a particular kind of rotational structure: you need to rotate it by seven hundred and twenty degrees — two full turns — to get back to the starting configuration. This sounds bizarre if you think of spin as literal rotation of a little ball, but it is perfectly natural for certain topological configurations. It is a concrete geometric property of the substrate knot, not an abstract quantum number that just happens to exist.

Mass is different from the other quantum numbers. It is not topological but dynamical. The mass of a particle is the energy cost of maintaining its localized configuration against the substrate's tendency to relax to smoothness. A tightly wound knot with steep gradients stores more energy in those gradients than a gently twisted one. More energy means more mass. This is why the top quark, presumably a very tightly wound and complex configuration, is so much heavier than the electron, which is presumably a simpler knot. And massless particles — photons, gluons — are not localized knots at all. They are

propagating disturbances, waves in the substrate that travel without being topologically trapped. They are not knots. They are ripples.

The stability and decay of particles also become transparent. An absolutely stable particle — the electron, the proton, the photon — is the lowest-energy state within its topological class. It cannot decay because there is no lighter configuration with the same topology. It would have to untie its knot, and topology forbids it. An unstable particle — the muon, the tau, the W and Z bosons, the Higgs — is topologically equivalent to some lighter combination of particles. Its knot can be rearranged, untied into simpler knots. The particle decays — not because it is pushed to decay by some external force, but because its own topology permits it, and thermodynamics (the tendency toward maximum entropy) drives it. Conservation laws — conservation of charge, conservation of lepton number, conservation of baryon number — are simply the conservation of topological invariants. You cannot change the total winding number in a local interaction. The topology is preserved.

The entire particle spectrum — the zoo of quarks, leptons, gauge bosons, and the Higgs — corresponds to the catalog of topologically distinct configurations the substrate supports. Three generations of fermions — the electron, muon, and tau, with their corresponding neutrinos, and similarly for the quarks — would be explained if the internal topology of the substrate supports exactly three families of knot types. The gauge bosons — photons, gluons, W and Z bosons — are propagating disturbances, not localized defects. The Higgs field is a condensate, a space-filling configuration of the substrate that breaks the electroweak symmetry and gives mass to the W and Z bosons. The framework provides a structural program: clear mathematical questions

whose answers determine the spectrum. How many topologically distinct knot types does the substrate's internal space support? That number should be the number of particle species. If it turns out to be exactly what we observe, the framework is vindicated. If not, it is falsified.

Now let me tell you about one of the deepest and most beautiful aspects of this framework: the way symmetries work.

In conventional physics, we start with symmetries. We observe that the laws of physics look the same regardless of your velocity (Lorentz invariance). We observe that they look the same regardless of your position or orientation (translational and rotational invariance). We observe that the fundamental forces respect certain internal symmetries described by mathematical groups —  $SU(3)$  for the strong force,  $SU(2)$  for the weak force,  $U(1)$  for electromagnetism. And then we build our theories to be compatible with these symmetries. The symmetries are inputs — fundamental rules that constrain what theories are allowed.

The substrate framework inverts this logic entirely. The symmetries are not inputs. They are outputs. The substrate has its own deeper symmetry — some fundamental symmetry group that we have not yet fully identified. When the substrate is coarse-grained and its collective behavior is examined at low energies, the familiar symmetries of physics *emerge*. Lorentz invariance emerges because the substrate's large-scale dynamics happen to respect it. Gauge symmetries emerge because the topology of the substrate's internal space happens to produce them. Diffeomorphism invariance — the symmetry that says the laws of gravity look the same in any coordinate system — emerges because the coarse-grained geometry is coordinate-independent.

This is a profound shift in perspective. Instead of asking, "What theory is compatible with the symmetries we observe?" the substrate framework asks, "What symmetries does the substrate's collective behavior naturally produce?" And it provides a reason *why* these particular symmetries exist. They are not arbitrary choices. They are inevitable consequences of the substrate's structure and dynamics.

All of the dynamics, by the way, come from a single action functional. One mathematical object encodes everything. It encodes the substrate's evolution. It encodes the emergence of spacetime. It encodes the behavior of particles. It encodes gravity. It encodes quantum mechanics. There is no separate action for gravity and another for matter. The Einstein-Hilbert action — the mathematical expression that generates Einstein's equations — is a low-energy limit, an approximation. The Standard Model Lagrangian — the mathematical expression that generates the equations of particle physics — is another low-energy limit. Both emerge from the one substrate action when you coarse-grain and take appropriate limits. The action must be unitary (no collapse, no information loss), complete (no separate sectors for different forces), finite (no infinities that need renormalizing — all divergences are artifacts of the coarse-graining), and background-independent (no pre-existing spacetime assumed — spacetime is derived from the action, not put in by hand).

Now I want to tell you about black holes, because the substrate framework resolves some of the most dramatic and long-standing puzzles about them.

Start with the information paradox. In 1975, Stephen Hawking made one of the most remarkable discoveries in theoretical physics: black holes radiate. They are not perfectly black. Quantum effects near the event horizon cause

them to emit particles — very slowly, very faintly — and over immensely long timescales, a black hole will evaporate completely. The problem is, Hawking showed that this radiation is thermal — it is random, featureless, carrying no information about what fell into the black hole. If a black hole forms from the collapse of a star, all the detailed information about that star — its composition, its structure, its history — is apparently lost when the black hole evaporates. The information just vanishes.

But that violates a sacred principle of quantum mechanics: unitarity. The principle that quantum evolution is reversible, that information is never truly destroyed. If you know the complete quantum state of the universe at any moment, you can in principle reconstruct the state at any other moment, past or future. Information loss would destroy this reversibility. It would mean quantum mechanics is wrong at a fundamental level.

For forty years, this paradox has been one of the deepest puzzles in physics. Is Hawking right, and information really is lost? Or is there some subtle mechanism that preserves the information and smuggles it out in the radiation?

The substrate framework resolves this cleanly. Evolution is strictly unitary. Information is never destroyed. What happens is that the Hawking radiation is *not* truly thermal. It *appears* thermal to an exterior observer because the observer has traced out — has no access to — the substrate degrees of freedom inside the horizon. The apparent randomness of the radiation is not a fundamental property. It is a consequence of limited access. The radiation actually carries subtle correlations — extremely delicate quantum correlations that are nearly impossible to detect individually but that collectively encode all the information about what fell into the black hole. As the black hole

evaporates, the information gradually transfers from the interior substrate configurations to correlations among the emitted radiation. The final state, after complete evaporation, is a pure quantum state — not a mixed, information-free thermal state. Nothing is lost. Nothing is destroyed. Information is redistributed, spread across correlations that are, in practice, inaccessible to local observations, but are there in principle.

And the event horizon — that supposedly impermeable barrier — is an emergent feature of the geometry. At the substrate level, there is no sharp boundary, no absolute barrier. The substrate is continuous across the horizon. Degrees of freedom on either side are connected, entangled, part of the same continuous fabric. The event horizon is where the emergent geometric description says "you cannot escape," but the underlying substrate has no such prohibition on information flow. It is like the speed of sound in a fluid: a surface where the flow speed exceeds the sound speed looks like a "sonic horizon" from which sound cannot escape, but information can still travel through the fluid itself, through modes other than sound. The event horizon is a limitation of the geometric description, not a feature of the fundamental reality.

Now let me tell you about an absolutely beautiful conjecture that the substrate framework turns into a theorem. In 2013, Juan Maldacena and Leonard Susskind proposed what they called ER equals EPR. ER stands for Einstein-Rosen bridge — a wormhole, a tunnel connecting two distant regions of spacetime. EPR stands for Einstein-Podolsky-Rosen — the initials behind the famous thought experiment about quantum entanglement. Maldacena and Susskind conjectured that these two phenomena — wormholes and entanglement — are actually the same thing. Every pair of entangled particles



is connected by a wormhole. Every wormhole is a manifestation of entanglement.

In the substrate framework, this is not a conjecture. It is a consequence. Entangled substrate configurations share direct connections at the substrate level — they are linked in the network of substrate degrees of freedom. When you coarse-grain and produce an emergent geometry, a direct substrate connection between two separated regions manifests as a geometric bridge — a wormhole. The width of the wormhole corresponds to the amount of entanglement. Maximum entanglement produces a substantial bridge. Less entanglement produces a narrower one. No entanglement produces no bridge at all — the regions are geometrically disconnected.

And this means that "spooky action at a distance" — the seemingly instantaneous correlation between entangled particles — is not really action at a distance at all. The entangled particles are not truly distant. They are connected through the substrate. They look distant in the emergent geometry, just as two cities might look distant on a map even though they are connected by an underground tunnel. The correlation does not have to travel through emergent spacetime. It is built into the substrate connectivity.

There is one more black hole puzzle I want to tell you about, because it beautifully illustrates the power of the substrate approach. In 2012, a group of physicists — Ahmed Almheiri, Donald Marolf, Joseph Polchinski, and James Sully, known by the acronym AMPS — showed that three assumptions physicists had long considered sacrosanct are actually mutually inconsistent near a black hole's event horizon. The three assumptions are: unitarity (information is preserved), the equivalence principle (an observer crossing the horizon notices nothing special), and quantum field theory on curved

spacetime (the standard framework for doing quantum physics near a black hole). AMPS showed that the monogamy of entanglement — the principle that a quantum system can be maximally entangled with only one other system — forces a contradiction. The Hawking radiation must be entangled with the early radiation (to preserve information) and simultaneously entangled with the interior (to ensure the equivalence principle). But monogamy says it cannot be maximally entangled with both. Something has to give. AMPS suggested that what gives is the equivalence principle: there must be a "firewall" of high-energy particles at the horizon, destroying anything that tries to cross. This was alarming. The equivalence principle — the foundation of general relativity — apparently fails at horizons.

The substrate framework resolves this by pointing out that the three assumptions of AMPS are not independent. The interior and exterior of the black hole are not independent quantum systems. They are regions of one substrate with continuously connected degrees of freedom. The quantum state does not factorize cleanly into "inside" and "outside" and "radiation." In the substrate, these are not three separate things — they are three overlapping descriptions of one interconnected system. The apparent monogamy violation arises only if you insist on treating them as independent, which the substrate framework says is illegitimate. The substrate is continuous across the horizon. The degrees of freedom are shared. There is no firewall because there is no sharp boundary at the fundamental level. The equivalence principle is preserved. Unitarity is preserved. And no new postulates or exotic physics are needed.

Let me now tell you about the predictions, because a framework that makes no predictions is philosophy, not physics. And this framework makes

predictions. Specific, quantitative, testable predictions — more than a dozen of them — several of which can be tested with experiments and observatories that already exist or are currently under construction.

First, and perhaps most striking: if spacetime has a discrete microstructure at the Planck scale, then light should travel at very slightly different speeds depending on its energy. Higher-energy photons interact more strongly with the microstructure and should travel at a slightly different speed than lower-energy photons. The deviation would be fantastically small — proportional to the ratio of the photon's energy to the Planck energy, which is tiny for any individual photon. But gamma-ray bursts — colossal explosions billions of light-years away — produce photons spanning an enormous range of energies, and those photons travel for billions of years. Over such immense distances, even a tiny speed difference accumulates. Higher-energy photons could arrive milliseconds or even seconds before or after lower-energy ones. The Fermi Gamma-Ray Space Telescope is already looking for exactly this effect, and future observatories will push the sensitivity further.

Along similar lines, cosmic rays — the ultra-high-energy particles that rain down on Earth from deep space — should show a subtle shift in what is called the GZK cutoff, the theoretical maximum energy a cosmic ray can have. The substrate's microstructure should shift this cutoff by a tiny but potentially measurable amount. The Pierre Auger Observatory in Argentina, which monitors cosmic rays over an area the size of Rhode Island, is the facility to look for this.

Then there are gravitational wave signatures. When two black holes merge, the standard prediction from general relativity is a specific pattern of gravitational waves — a chirp that rises in frequency and amplitude, then rings

down. But if the substrate framework is correct and the event horizon has internal structure, there should be echoes — faint repetitions of the ringdown signal, arriving tens to hundreds of milliseconds after the main signal, produced by waves bouncing off the substrate structure near the would-be horizon. LIGO, Virgo, and KAGRA are currently sensitive enough to search for these echoes, and there have already been tentative (though not yet conclusive) reports of such signals in the data.

At higher frequencies, gravitational waves should also show dispersion — different frequencies traveling at slightly different speeds — similar to the light speed variation but for gravity waves. This would require future detectors sensitive to megahertz and gigahertz gravitational waves, which are currently in the conceptual design phase but are technologically feasible.

One particularly elegant prediction concerns gravitational wave polarization. General relativity allows gravitational waves to have two polarization modes — two ways the wave can stretch and squeeze space as it passes. The substrate framework predicts additional modes: scalar and vector polarizations beyond the two tensor modes of standard general relativity. Pulsar timing arrays — networks of millisecond pulsars that act as galaxy-sized gravitational wave detectors — can look for these extra polarization modes. The NANOGrav collaboration, which already detects a stochastic gravitational wave background, could potentially identify them.

The cosmic microwave background — the afterglow of the Big Bang, the oldest light in the universe — should carry subtle imprints of Planck-scale physics. The quantum fluctuations that seeded the formation of galaxies originated at incredibly small scales during inflation, and the substrate's microstructure should leave tiny corrections in the pattern of temperature and

polarization fluctuations. These corrections are below the sensitivity of current experiments but within reach of future missions like CMB-S4, LiteBIRD, and PICO, which are scheduled for the coming decade.

Here is a prediction that touches on one of the biggest mysteries in cosmology: dark energy. The universe is expanding, and the expansion is accelerating. Something is pushing space apart. In the standard picture, this is attributed to the cosmological constant — a constant energy density inherent to empty space. But there is an enormous problem: when physicists try to calculate the cosmological constant from quantum field theory, they get a number that is ten to the one hundred and twentieth power times larger than what is observed. This is the worst prediction in the history of science.

The substrate framework offers a different perspective. Dark energy is the residual tension of the substrate — energy stored in the fact that the substrate has not yet fully relaxed to its ground state after the phase transition that created spacetime. It is like a stretched rubber sheet that is slowly releasing its tension. This predicts that the dark energy equation of state — a parameter called  $w$  that describes how dark energy behaves — should be slightly greater than negative one. Standard cosmological constant gives exactly negative one. A substrate that is slowly relaxing gives something a tiny bit above that. The Dark Energy Spectroscopic Instrument, the Euclid space telescope, and the Nancy Grace Roman Space Telescope are all designed to measure  $w$  with sufficient precision to detect this deviation if it exists.

The primordial gravitational wave spectrum — the pattern of gravitational waves left over from the earliest moments of the universe — should carry a characteristic feature: a bump or suppression at a frequency corresponding to the phase transition scale where geometric spacetime condensed from the

pre-geometric substrate. This is a smoking-gun signature that future space-based gravitational wave observatories like LISA could detect.

One of the most exciting near-term predictions involves gravitational decoherence. If gravity is emergent from the substrate, then objects in quantum superposition should experience a very specific type of decoherence — a loss of quantum coherence — at a rate determined by the substrate's properties. For objects with masses around ten to the negative fourteen kilograms (much larger than atoms but much smaller than everyday objects) held in superposition with a separation of about one micrometer, the predicted decoherence time is between a thousandth of a second and a tenth of a second. Several proposed experiments — MAQRO, OTIMA, and various levitated nanoparticle experiments — are specifically designed to test this regime. This prediction is remarkable because it gives a specific number for a specific experimental setup, which is far more than most quantum gravity proposals can offer.

The generalized uncertainty principle — the modification of the Heisenberg uncertainty relation at very high energies due to the substrate's discrete structure — should produce corrections to the energy levels of heavy muonic atoms. These are atoms where a muon replaces an electron, and because muons are much heavier than electrons, they orbit much closer to the nucleus, probing much smaller distance scales. Experiments at CERN — ALPHA and ASACUSA — are sensitive enough to search for these corrections.

Black hole radiation, if we could observe it, should be non-thermal — it should contain subtle correlations that encode information, distinguishing it from purely random thermal radiation. While astrophysical black holes are far too large and cold for this to be directly observable, analogue black hole

experiments — laboratory setups that mimic the physics of black hole horizons using flowing fluids, Bose-Einstein condensates, or optical systems — can test the prediction that Hawking radiation carries information.

The shape of a black hole's shadow — the dark silhouette visible when a black hole is backlit by glowing gas — should differ from the general relativistic prediction by a tiny fraction, due to the substrate structure near the horizon. The Event Horizon Telescope, which has already produced stunning images of the black holes in M87 and at the center of our own galaxy, could potentially detect these deviations, especially with future upgrades that include space-based radio dishes for sharper resolution.

And there is one more prediction that I find particularly fascinating. It involves an experiment called the BMV experiment — named after Bose, Marletto, and Vedral, who proposed it. The idea is to take two small masses, put them both in quantum superposition, and see whether gravity can entangle them. If gravity is truly quantum and mediated by gravitons, it should produce entanglement at a specific rate. If gravity is emergent from the substrate, it should also produce entanglement, but at a slightly different rate. The difference is small but potentially measurable. This experiment is currently being developed by several groups around the world, and it represents perhaps the most direct way to distinguish between graviton-based and substrate-based pictures of gravity.

Now, I want to be honest with you about how this framework compares to the other approaches I mentioned earlier, because fairness matters and intellectual honesty matters, and a framework that only looks good when you ignore the competition is not worth much.

Compared to string theory, the substrate framework is dramatically more economical. There are no extra dimensions — the substrate produces three-plus-one dimensions from its own dynamics. There is no landscape of ten to the five hundredth possible universes — there is one substrate, one action, one universe. There are no new fundamental objects — no strings, no branes, no supersymmetric partners. Everything is the substrate, in different configurations. The framework extracts the physical content — unification, quantum gravity, a spectrum of particles — without the metaphysical baggage. Now, string theory has produced genuine insights, especially in mathematics and in the holographic principle, and many of those insights are actually compatible with the substrate approach. But as a physical theory of our specific universe, the substrate framework is more constrained and more directly falsifiable.

Compared to loop quantum gravity, the differences are equally sharp. Loop quantum gravity quantizes geometry — it starts with geometry and applies quantum rules to it. The substrate framework does not quantize geometry. It derives geometry from something deeper. Loop quantum gravity adds matter separately — particles and fields are placed onto the quantized spacetime mesh. The substrate framework derives matter and geometry from the same source. In loop quantum gravity, the hard problem is recovering smooth spacetime at large scales. In the substrate framework, that recovery is built into the coarse-graining procedure from the start.

Compared to causal set theory, the substrate framework does not commit to any specific discrete structure. Causal sets assume spacetime is fundamentally a discrete set of events with causal relations. The substrate framework says the substrate has its own structure, and discreteness is one of



its features, but the specific form of that structure is determined by the dynamics of the action, not assumed in advance.

Compared to emergent gravity proposals — ideas that gravity might emerge from thermodynamics or information — the substrate framework provides what those proposals have been missing: a microscopic foundation. Emergent gravity proposals like Erik Verlinde's entropic gravity or Jacobson's thermodynamic derivation are beautiful but incomplete. They show that Einstein's equations can be derived from thermodynamic reasoning, but they do not tell you what the microscopic degrees of freedom are, or what their dynamics are, or why entropy should be proportional to area. The substrate framework fills these gaps.

And compared to John Wheeler's famous slogan "it from bit" — the idea that physical reality emerges from information — the substrate framework makes the metaphor concrete. The substrate is the physical system whose information content is the "bits." The emergence of spacetime and matter from the substrate is the "it from bit" program realized in a specific, mathematical, testable framework.

But let me also tell you what we do not know. Because a good theory is one that tells you clearly what it has accomplished and what it has not, and the substrate framework has significant open problems.

The most important open problem is the specific form of the substrate action. The framework tells you that there is a single action functional that governs everything. It tells you the properties this action must have — unitarity, completeness, finiteness, background independence. But it does not yet tell you what the action actually is. This is the central mathematical

challenge. It is like knowing that there must be an equation governing a system and knowing what properties the equation must have, but not yet knowing the equation itself. Until the action is found, many of the framework's qualitative predictions cannot be made quantitative, and the program remains, in a sense, a promissory note. A very detailed, very specific promissory note — but a promissory note nonetheless.

The complete derivation of the Standard Model particle spectrum from the topology of the substrate has not been carried out. The framework provides a clear program — classify the topologically distinct configurations — but actually doing the classification requires knowing the internal topology of the substrate, which requires knowing the action. Similarly, calculating the cosmological constant from the substrate requires knowing the action. The computational methods for working with pre-geometric degrees of freedom — degrees of freedom that do not live in space and time — are still being developed. And the rigorous derivation of the classical limit — showing in full mathematical detail that smooth spacetime and Newtonian gravity emerge from the substrate in the appropriate regime — has not been completed to the standards of mathematical physics.

There are also things the framework does not explain and does not claim to explain. It does not explain why the substrate exists. Why is there something rather than nothing? That is a metaphysical question, not a physical one, and the substrate framework, like every physical theory, takes the existence of its fundamental entity as a starting point. It does not explain the specific values of the constants of nature — why Planck's constant has the value it has, why the fine-structure constant is approximately one over one hundred and thirty-seven — although it reframes these as questions about the

substrate's structure rather than irreducible mysteries. And it does not explain consciousness, which some might consider relevant to the measurement problem but which the framework addresses purely in terms of decoherence and information redistribution.

Despite these open problems, the framework has some remarkable advantages that I think are worth emphasizing.

Economy. One field. One action. No extra dimensions, no extra particles, no extra forces, no extra principles. Everything from one source. This is not just aesthetic elegance — it is explanatory power. The fewer free elements a theory has, the more constrained it is, and the harder it is to fudge.

Conceptual clarity. The framework resolves longstanding puzzles — the measurement problem, the information paradox, the firewall paradox, the origin of Bekenstein-Hawking entropy, the unification of gravity and quantum mechanics — not by adding new mechanisms but by removing false assumptions. Most of these puzzles arise from treating emergent features as fundamental. When you stop doing that, the puzzles dissolve.

Falsifiability. Thirteen specific predictions, several testable with current or near-future technology. This is a higher density of predictions per framework than any other quantum gravity program currently offers. Most quantum gravity approaches have struggled to make even a single prediction that could be tested within our lifetimes. The substrate framework gives you more than a dozen, with specific numbers, specific experiments, and specific timelines.

And natural unification. The framework does not unify gravity and quantum mechanics by forcing one onto the other — by quantizing gravity or by geometrizing quantum mechanics. It derives both from something deeper.

General relativity and quantum mechanics are both approximate descriptions, both valid in their respective regimes, both emerging from the substrate's dynamics in the appropriate limits. They do not need to be unified because they were never separate. They are two shadows cast by the same object, seen from different angles.

Let me close with some perspective. The history of physics is, in many ways, a history of unifications.

In the nineteenth century, James Clerk Maxwell showed that electricity and magnetism — two phenomena that seemed utterly different, one involving charges and sparks, the other involving magnets and compass needles — were actually two aspects of a single electromagnetic field. One field, two manifestations. It was one of the greatest intellectual achievements in human history, and it gave us radio, television, radar, wireless communication, and the entire modern world.

In the early twentieth century, Albert Einstein showed that space and time — two concepts that seemed utterly different, one the stage on which events occur, the other the sequence in which they happen — were actually two aspects of a single spacetime geometry. One geometry, two manifestations. It was another epochal unification, and it gave us our understanding of gravity, cosmology, black holes, and the expanding universe.

The substrate framework proposes the next step in this grand progression: matter and spacetime — the things that exist and the arena they exist in — are actually two aspects of a single substrate. One field, two manifestations. If this is correct, it would be a unification on the same level as Maxwell's and Einstein's — perhaps deeper, because it would subsume both of them.

Whether it is correct, of course, is not a question of elegance or philosophy or personal preference. It is a question of mathematics and experiment. Can the action be found? Can the Standard Model spectrum be derived? Can the predictions be confirmed or refuted? These are questions with answers, and the answers will come from the hard, unglamorous, magnificent work of calculation and observation.

The predictions have been stated. The experiments have been identified. The timelines are measured in years, not centuries. We will know, within a generation or two, whether the fabric beneath everything is one fabric or many — whether the deepest level of reality is a single substrate weaving spacetime and matter from the same thread, or whether the truth is something stranger still that no one has yet imagined.

The work begins now. And I think that is wonderfully exciting.